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2027 CYCLE

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CUET UG

Biology

Evolution

Official Syllabus Unit VII · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit VII)	Skills the questions test
Origin of life	Origin of life – Miller's experiment.	Spotting the Miller–Urey setup and its result; early-Earth conditions; the RNA-world idea and first-cell dates.
Concepts of evolution	Concepts of evolution (Darwin's contribution).	Stating what selection needs; branching descent; the peppered-moth story; the Origin of Species date.
Evidences for evolution	Evidences for evolution (embryology, palaeontology, comparative anatomy, divergent and convergent evolution and industrial melanism).	Sorting homologous from analogous organs and naming the evolution type each points to.
Adaptive radiation	Adaptive radiation.	Explaining one ancestor → many forms; matching the finch and marsupial examples.
Biological evolution	Biological evolution.	Ordering the big events: invertebrates, jawless fish, ferns, dinosaurs, mammals.
Natural selection	Natural selection.	Identifying stabilising, directional and disruptive selection from a description.
Mechanism of evolution	Mechanism of evolution – Hardy–Weinberg principle and affecting factors (gene migration, genetic drift, mutation, recombination and natural selection).	Computing p^2, $2pq$ and q^2; naming which of the five factors disturbs equilibrium.
Evolution through geological periods	Evolution of plants and vertebrates through geological periods.	Matching dates to events; reading the timeline; fossil logic (deeper rock = older).

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Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit VII)	Skills the questions test
Origin and evolution of man	Origin and evolution of man.	Ordering the human ladder; matching each species to its trait and date.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Biology paper maps strictly to NCERT Class 12, and Unit VII pairs genetics with its big payoff: how the variety genes create becomes the story of life's change over time.

Origin of Life: Chemistry Before Biology

Here is a puzzle you can hold in your hand. An apple falls from a tree, rots on the ground, and no apple tree sprouts from the rot. In our world, life always comes from life. So how did the **first** living thing appear, on a planet that had nothing alive on it at all? Biology's honest answer: the early Earth was a giant chemistry set, and under its conditions the building blocks of life could assemble on their own. This section walks that story forward, one date at a time.

From the Big Bang to a habitable Earth

- **The universe is old:** almost **20 billion years** — its matter and energy were squeezed into one point and exploded outward in the **Big Bang**, a single huge explosion astronomers use to explain the origin of the universe itself.
- **Gases came first:** as the explosion's heat dropped, hydrogen and helium formed, clumped by gravity, and built the first stars and galaxies.
- **Earth forms:** our planet condensed about **4.5 billion years ago** (4.5 bya), still molten and bubbling.
- **No oxygen at all:** the early air had **no oxygen gas** — instead, water vapour, CH_4 (methane), CO_2 (carbon dioxide) and NH_3 (ammonia) poured out of the molten crust, and ultraviolet rays from the Sun beat down freely.
- **Oceans boil over:** as Earth cooled, the water vapour condensed into clouds, rained for ages, and filled the hollows — an ocean rich in dissolved gases.

Condition	Early Earth (before life)	Earth today
Atmosphere	Water vapour, CH_4, CO_2, NH_3 — no O_2	78% nitrogen, 21% oxygen
Energy source	Lightning, ultraviolet rays, volcanic heat	Mostly sunlight, filtered by ozone

Condition	Early Earth (before life)	Earth today
Surface water	Hot, gas-saturated seas after long rains	Cool, oxygenated oceans
Living things	None — only chemistry	Millions of species everywhere

The Oparin–Haldane idea: life rose from non-life

- **Two thinkers, one idea:** in the 1920s, Russian chemist **Alexander Oparin** and British scientist **J.B.S. Haldane** proposed **chemical evolution** — life did not appear ready-made; it assembled **step by step** from non-living molecules.
- **The steps:** simple gases → simple organic molecules (sugars, amino acids) → bigger molecules (proteins, nucleic acids) → first cell-like packages.
- **Energy did the work:** with no oxygen to break things down, lightning and UV rays could drive chemistry that today's air would never allow.
- **Why it sounds strange:** we never see this happen now, because today's oxygen instantly attacks such molecules — the recipe only worked on the early planet.

WORKED EXAMPLE — MILLER'S 1953 EXPERIMENT, STEP BY STEP

Stanley Miller, working in Harold **Urey's** laboratory, tested the Oparin–Haldane idea directly in 1953.

- **Step 1:** he sealed a glass flask containing CH_4, H_2, NH_3 and water vapour — the reconstructed early atmosphere, with **no oxygen**.
- **Step 2:** he kept the mixture at about **800°C** and passed an **electric spark** through it — the spark stood in for lightning.
- **Step 3:** he let the mixture circulate, cool, and collect — as rain would have washed products into early seas.
- **Observation: amino acids** (the building blocks of proteins) formed from the gases — no cells, no enzymes, only chemistry and energy.
- **Conclusion:** the building blocks of life **can** assemble under early-Earth conditions. Later experiments of the same kind produced sugars, nitrogen bases, pigments and fats too.

The point students miss: Miller did **not** make life — he made life's **parts**, and that alone is powerful evidence for chemical evolution.

Other ideas people held

- **Panspermia:** early Greek thinkers suggested units of life (spores) travelled through space and seeded Earth — a possibility still discussed, but with no proof that life started anywhere but chemistry.
- **Spontaneous generation:** the old belief that life springs from rotting matter — mud makes frogs, straw makes mice. **Louis Pasteur** demolished it with swan-neck glassware: boiled broth stayed sterile unless dust (carrying microbes) reached it. Life comes **only** from pre-existing life — today.

- **Divine creation:** the belief that all life was created by a supernatural power — held by faith, but it makes no scientific predictions and cannot be tested.

ANALOGY — THE UNTENDED SOUP POT

Leave a pot of the right gases simmering under a spark-thrower, and new molecules keep forming in the broth — that was Miller's flask, and the early ocean was the same pot at planetary size. The mapping is honest: gases = ingredients, lightning and UV = the flame, the ocean = the pot, and the amino acids that form = the first trace of cooking. But the pot never "wants" anything — the recipe runs by chemistry alone, with nobody stirring.

The RNA world and the first cells

- **RNA came first:** the strongest evidence says **RNA** — a molecule that can both store information and catalyse reactions — was the **first genetic material**; essential processes such as metabolism and protein-making evolved around it.
- **Non-cellular, then cellular:** the earliest life was probably giant RNA-style molecules floating free; the **first cells** appeared around **3.8 billion years ago**, and all of them were **single-celled prokaryotes**.
- **Why RNA lost the job:** DNA is more stable than RNA, so DNA took over information storage and RNA kept the working roles — the arrangement every living cell still uses.

Darwin's Contribution: Natural Selection

In 1831 a 22-year-old naturist named Charles Darwin boarded a survey ship, HMS **Beagle**, for a five-year trip around the world. He came back with notebooks full of fossils, birds and questions — and one of the most powerful ideas in science. Darwin saw that every population carries endless small differences, and that nature quietly keeps the differences that help their owners survive and reproduce. Twenty-odd years later, that thought became the theory of evolution by natural selection.

The voyage that changed biology

- **HMS Beagle, 1831–1836:** Darwin sailed as the ship's naturalist, collecting specimens and fossils on every coast he visited.
- **The Galápagos finches:** on these volcanic islands he found many finch species — **seed-eaters with big beaks, insect-eaters with slender beaks**, even a woodpecker-like beak for probing — all clearly related.
- **The puzzle:** why would closely related birds differ island by island, in ways that exactly fit each island's food? His answer took decades of thinking.
- **Heritable variation is the raw material:** members of any population differ in tiny traits, and many of those traits pass from parent to offspring.

The core idea: natural selection

Darwin's mechanism needs four ingredients working together:

- **Tiny heritable variations:** individuals differ slightly, and the differences can be inherited by the next generation.

- **Struggle for existence:** food, mates and space are limited, while more offspring are produced than can possibly survive.
- **Survival and reproduction of the fittest:** "fitness" means **reproductive** fitness — leaving more surviving offspring — not being the strongest.
- **Branching descent:** over generations, selected traits accumulate, populations split, and all forms descend from shared ancestors — the tree of life branches.
- **The book:** Darwin published **On the Origin of Species** in **1859**, laying out the whole argument with evidence.
- **He was not alone: Alfred Russel Wallace**, working in the Malay Archipelago, reached the same conclusion and the two presented the idea together in **1858** — Wallace deserves that line of credit.

Industrial melanism: selection caught in the act

- **The moth: Biston betularia**, the peppered moth of England, flies at night and rests on tree trunks by day, where birds hunt it by sight.
- **Before industry:** trunks were pale, covered in lichen — the **white-winged** form was camouflaged, and dark moths were eaten more often.
- **After industry:** soot killed the lichen and darkened the trunks — now the **dark-winged** form was hidden, and it became the common one.
- **What it proves:** predators (the birds) **selected** the better-camouflaged colour in each setting. The environment changed, the selection pressure flipped, and the moth population's appearance followed — evolution in action.

Stage	Tree trunks	Moth form that survives
Pre-industrial countryside	Pale, lichen-covered bark	White-winged moths — dark ones are spotted and eaten
Industrial, soot-covered city	Dark, bare bark	Dark-winged moths — white ones stand out

Selection you can watch today

- **Antibiotic-resistant bacteria:** a course of antibiotics kills the sensitive cells; the few resistant cells survive, reproduce, and soon the whole infection is resistant — selection inside a patient's body.
- **Pesticide-resistant pests:** spraying kills the susceptible insects; resistant ones multiply, so each new chemical fails faster than the last.
- **Herbicide-resistant weeds:** the same story in farm fields — weeds that shrug off a herbicide spread through every treated field.
- **The pattern is identical:** a pressure kills the unprotected, survivors pass on their protection, and the population's average shifts.

Three shapes of selection

Mode	What selection favours	Classic example
Stabilising	The average; both extremes are removed	Human birth weight — very small and very large babies are at greatest risk
Directional	One extreme, so the average shifts that way	A drought leaves only large hard seeds — finches with bigger beaks survive and the average beak grows
Disruptive	Both extremes at once, against the middle	A plant with only very large or very small seeds — birds ignore extremes but eat the mid-sized ones, so mid-sized beaks lose

ANALOGY — THE GARDENER AND THE WEEDS

A gardener pulls every plant that lacks the traits she wants and lets the rest seed — that is **artificial selection**, humans choosing by design. Natural selection is the same sieve with no gardener: the environment (drought, soot, antibiotics, birds' eyes) removes whatever does less well, and the survivors do the reproducing. The mapping to state honestly: the sieve = the struggle for existence, the kept seeds = the fittest, and the next year's garden = the next generation, already shifted toward what survived.

WORKED EXAMPLE — READING A SELECTION MODE FROM A DESCRIPTION

Statement: "A population of fish lives in a lake with predators that hunt in both very shallow and very deep water. Over generations, fish are mostly small (hiding near plants) or large (too big to swallow); medium-sized fish are the rarest."

- **Clue 1: both** extremes survive best → not directional.
- **Clue 2:** the middle group is the one being removed → this is **disruptive** selection.
- **Check:** one extreme? → directional. Only the average kept? → stabilising. Both extremes kept, middle cut? → disruptive.

Three descriptions, three answers — train the reflex of asking **who is being removed**, and the mode names itself.

Evidences for Evolution

If life evolves, the evidence should be lying around everywhere — in rocks, in eggs, in our own arms. It is. Scientists read evolution from four big sources: the fossil record, embryos, the anatomy of living things, and selection seen happening in front of us. Each source gives a different kind of

clue, and questions love to make you sort examples into the right bin. Learn the bins first; the examples will then file themselves.

Palaeontology: the fossil record

- **What a fossil is:** any preserved remains or impression of an organism — bones, shells, leaf prints — trapped in rock layers (strata) of the earth.
- **Deeper means older:** rock layers form one on top of another, so a fossil in a deep layer lived earlier than a similar fossil near the surface.
- **Missing links:** some fossils sit halfway between two big groups — **Archaeopteryx** shows a reptile-like toothed jaw, long tail and claws **plus** feathered wings, marking the step from reptiles toward birds.
- **Ammonites and dinosaurs:** whole sequences of species appear, change and vanish through the layers — life's history written in stone.
- **How ages are fixed:** fossils' ages are estimated by **dating** the rocks around them, which strengthens the deeper = older logic.

Embryology: shared beginnings

- **The observation:** early embryos of fish, salamander, tortoise, chick, rabbit and human look strikingly alike — all show **gill pouches** just behind the head and a **tail**, whatever they will become.
- **What it means:** gill pouches make perfect sense in a fish, but why would a human embryo build them? Because vertebrates inherited the same developmental plan from a common ancestor.
- **Von Baer's correction:** Karl Ernst von Baer dismissed the old slogan "ontogeny recapitulates phylogeny" (the idea that an embryo replays its ancestors' adult stages) — embryos never pass through the **adult** stages of other animals; they simply share early **embryonic** features.
- **The safe sentence:** similar embryos point to common ancestry — they do not "become" each other.

Comparative anatomy: same parts, different jobs

- **Homologous structures:** organs in different animals that share the **same basic structure and origin** but perform **different functions**.
- **The forelimb family:** the flipper of a whale (swimming), the wing of a bat (flying), the foreleg of a cheetah (running) and the arm of a human (grasping) all run on the same bone plan — humerus, radius-ulna, carpals.
- **A plant version:** the **thorn of Bougainvillea** and the **tendril of Cucurbita** are both modified branches — same origin, different work (defence versus climbing support).
- **Divergent evolution:** this pattern — one ancestral structure, modified for different needs — is called divergent evolution, and it is the signature of **common ancestry**.

Same job, different parts

- **Analogous structures:** organs with **completely different structures and origins** that have been shaped to do the **same function**.

- **Wings:** a butterfly's wing is a flat outgrowth of its body wall; a bird's wing is a modified forelimb with bones — same job (flight), no shared plan.
- **Eyes:** the eye of an octopus and the eye of a mammal look alike and both focus images, yet they developed from entirely different embryonic tissues.
- **Flippers:** penguin and dolphin flippers both drive swimming, but one is a bird's wing and the other a mammal's forelimb.
- **A plant pair:** sweet potato (a root) and potato (a stem) both store food — different organs, same function.
- **Convergent evolution:** this pattern — unrelated lines pressed by similar needs into similar tools — is called convergent evolution; it is the signature of **similar habitats**, not of close kinship.

Feature	Homologous organs	Analogous organs
Basic structure and origin	Same — inherited from a common ancestor	Different — built from unrelated parts
Function	Different (adapted to new needs)	Same (solves the same problem)
Evolution type	Divergent evolution	Convergent evolution
Animal example	Whale flipper, bat wing, cheetah leg, human arm	Butterfly wing and bird wing; octopus eye and mammal eye; penguin and dolphin flippers
Plant example	Thorn of Bougainvillea and tendril of Cucurbita	Sweet potato (root) and potato (stem) — both store food

Industrial melanism, one more time

- **The recap:** the peppered-moth shift belongs in this list too — it is **ecological evidence**, showing selection producing change within a human lifetime, exactly as the fossil record shows it across millions of years.

WORKED EXAMPLE — FOUR PAIRS, FOUR VERDICTS

Classify each pair as homologous or analogous, and name the evolution type:

- **Whale flipper vs bat wing:** same forelimb bones, different jobs → **homologous → divergent evolution**.
- **Butterfly wing vs bird wing:** unrelated structures, same job (flight) → **analogous → convergent evolution**.
- **Thorn of Bougainvillea vs tendril of Cucurbita:** both modified branches → **homologous → divergent evolution**.
- **Sweet potato vs potato:** root vs stem, both food stores → **analogous → convergent evolution**.

The two-question test: **same plan?** → homology. **same job?** → analogy. Answer the plan question before the job question and you cannot be trapped.

Adaptive Radiation: One Ancestor, Many Forms

Drop a few colonists on an empty island chain and something remarkable happens: their descendants split into many species, each shaped for a different way of life. This burst of branching from a single starting stock is **adaptive radiation** — the evolution of different species in one geographical area, starting from a point and literally radiating into other habitats and niches (a **niche** is a species' "job description": what it eats, where it lives, how it makes its living).

The idea in plain words

- **One ancestor:** all the new forms descend from a single original population.
- **Many openings:** the new area holds empty niches — no competitors yet.
- **Fast branching:** variants that fit each niche survive and reproduce there, so the population splits into specialised forms — a branching sprint of evolution.

Darwin's finches: the textbook case

- **The starting stock:** one seed-eating ground finch from South America reached the Galápagos Islands long ago.
- **The radiation:** its descendants evolved many beak shapes — stout beaks for seeds, slender insect-eating beaks, and a woodpecker-like beak for probing and picking insects out of bark.
- **Why beaks?** The beak is the finch's food tool, so each food niche selected the beak that worked best — the same logic as the moth, played out into separate species.
- **The lesson:** closely related birds on neighbouring islands differ **because** they radiated, not by accident of creation.

Australian marsupials: the parallel radiation

- **The setup:** Australia drifted away with only marsupials (pouched mammals) on board; their descendants radiated into the same niches that placental mammals — mammals whose young develop fully inside the mother — filled elsewhere.
- **The result:** for almost every placental mammal on other continents, Australia grew a marsupial look-alike that lives the same life.

Niche	Australian marsupial	Placental counterpart elsewhere
Small ground hunter	Marsupial mouse	Mouse
Burrowing digger	Marsupial mole	Mole
Large predator	Marsupial wolf (thylacine)	Wolf

- **Reading the table:** the pairs are **not** close relatives — they are separate radiations that converged on the same lifestyles, which is why marsupial and placental look-alikes are a classic convergent-evolution pairing.
- **Two patterns, one family tree:** radiation fills **new areas** from one stock (finches); convergence reshapes **different stocks** in similar niches (marsupials vs placentals). Both leave the same message: environments select.

ANALOGY — ONE TRAINING BATCH, MANY DEPARTMENTS

Imagine one training batch joining a brand-new company with every department empty. One member ends up in sales, one in accounts, one in the workshop — same starting skills, different jobs, and after years they barely resemble each other. The batch is the common ancestor, the empty departments are the unfilled niches, and the specialists they become are the radiated species. The mapping stays honest because nothing in the story invents new skeletons — it only redeploys what the family already had.

The Mechanism: Hardy–Weinberg and the Five Factors

So far you have seen **that** populations change. Now comes the part of Unit VII where biology meets arithmetic: a small equation that lets you **measure** whether a population is evolving. The whole game is built on one observation — a population is a **pool of alleles**, and you can count what fraction of the pool each version of a gene holds. If that fraction is not changing, the population is standing still; if it is changing, evolution is happening.

Allele frequencies: counting the pool

- **Allele:** one version of a gene — say **A** (dominant form of a trait) or **a** (its recessive partner).
- **Frequency:** the **fraction** of all copies of that gene in the population that are A or a — for example, if 70% of the copies are A, then $p = 0.7$.
- **Genotype, not just allele:** individuals pair the alleles up — AA, Aa or aa — and the frequencies of these three genotypes must add up to 1.

THE HARDY–WEINBERG EQUATION

$p^2 + 2pq + q^2 = 1$

$p + q = 1$

- p^2 = frequency of **AA** (homozygous dominant)
- $2pq$ = frequency of **Aa** (heterozygous)
- q^2 = frequency of **aa** (homozygous recessive)
- p = frequency of allele A; q = frequency of allele a

- **The principle, in words:** in a **large** population that mates **at random**, with **no evolutionary forces** acting, these frequencies stay identical generation after generation — a state called **genetic equilibrium**.
- **The flip side: evolution is exactly a change in allele frequency** between generations. Disturb the equilibrium and you are watching evolution happen.

WORKED EXAMPLE — FORWARD: FREQUENCIES FROM ALLELES

In a population, the allele frequencies are $p = 0.7$ and $q = 0.3$. Find the genotype mix.

- **Step 1 — homozygous dominant:** $p^2 = (0.7)^2 = 0.49 \rightarrow$ **49% are AA.**

- **Step 2 — heterozygous:** $2pq = 2 \times 0.7 \times 0.3 = 0.42 \rightarrow$ **42% are Aa**.
- **Step 3 — homozygous recessive:** $q^2 = (0.3)^2 = 0.09 \rightarrow$ **9% are aa**.
- **Check:** $0.49 + 0.42 + 0.09 = 1.00$, and $0.7 + 0.3 = 1.0$ — the population's gene pool is fully accounted for.

WORKED EXAMPLE — REVERSE: ALLELES FROM A COUNT

A survey finds that **16% of individuals** show a recessive trait (aa). Find the heterozygote frequency.

- **Step 1:** $q^2 = 0.16$, so $q = \sqrt{0.16} = 0.4$.
- **Step 2:** $p = 1 - q = 1 - 0.4 = 0.6$.
- **Step 3:** heterozygotes $= 2pq = 2 \times 0.6 \times 0.4 = 0.48 \rightarrow$ **48% are Aa**.
- **Sanity note:** carriers (48%) are far more common than affected individuals (16%) — recessive alleles mostly hide inside heterozygotes.

The five factors that disturb equilibrium

Hardy–Weinberg balance is a **null model** — it holds only when nothing is pushing. Five forces push, and each one is a named syllabus pointer:

Factor	What it does to the pool	Classic situation
Gene migration (gene flow)	Individuals move in or out, carrying their alleles between populations	Migrants from a neighbouring herd join and add new alleles
Genetic drift	Chance — not fitness — changes frequencies, strongest in **small** populations	A few founders start an island population; a crash leaves random survivors (bottleneck)
Mutation	New alleles appear, or one allele turns into another	A copying error creates a version no individual carried before
Recombination	Crossing over shuffles existing alleles into new combinations each generation	New trait combinations appear in offspring without new alleles
Natural selection	Alleles that improve survival or reproduction become more common	Soot-dark trees raise the dark-moth allele generation after generation

- **Drift's two famous faces:** the **founder effect** (a new population started by a few individuals carries an unrepresentative allele mix) and the **bottleneck** (a population crash leaves survivors whose alleles were picked by luck).

- **Why size matters so much:** in a small population, one random death or one lucky mating swings the frequencies — chance has room to steer; in a large population, chance events cancel out.
- **The exam reflex:** a question that says "random change in a small population" means **genetic drift**; "individuals entering a population" means **gene migration**; "new allele appears" means **mutation**.

Life Through the Ages: Plants and Vertebrates

Time is the hardest thing in biology to feel. Our whole recorded history is a few thousand years, yet evolution plays its game across **billions**. Squeeze all 4.5 billion years into a single calendar year and you get the right shiver: dinosaurs vanish around 26 December, "Lucy" appears on the evening of 31 December, and farming begins in the last minute before midnight. The walk below keeps the standard markers the syllabus and NCERT use — learn the order first, the dates second.



Figure 1 — Life through the ages: twelve milestones from the formation of Earth to the spread of Homo sapiens, with dates as given in the standard geological story.

The walk, marker by marker

- **2000 million years ago (mya): invertebrates** — animals without backbones — flourished in the seas, and cellular life (all of it single-celled prokaryotes) was already established.
- **500 mya: jawless fish** evolved — the first animals with a backbone.
- **350 mya: giant ferns** (pteridophytes) formed swampy forests on land; some lobe-finned fishes — fish with limb-like fins and lung-like organs — gave rise to the first **amphibians**, which could live both in water and on land.
- **The land conquest continues:** amphibians evolved into **reptiles**, which lay thick-shelled eggs that do not dry out — so reptiles spread far inland, while seed ferns gave way to **gymnosperms** (cone-bearing, naked-seed plants).
- **230 mya:** the **dinosaurs** — reptiles — rose and dominated the land.

- **65 mya:** dinosaurs vanished **suddenly** (the leading suspects are climate change and an asteroid impact); the surviving **mammals** spread into every empty niche and flourished.
- **A living fossil:** in 1938 a **coelacanth** — a lobe-finned fish thought to have been extinct for millions of years — was caught off South Africa, a live reminder of the fish lineage that walked toward land.

When	Animals	Plants and habitats
2000 mya	Invertebrates flourish in the seas	Ocean-dwelling life; land still bare
500 mya	Jawless fish — first backboned animals	Seas rich in algae
350 mya	Lobe-finned fish → first amphibians walk on land	Giant ferns (pteridophytes) in swampy forests
230 mya	Reptiles rule; dinosaurs dominate	Gymnosperms (conifers) widespread
65 mya	Dinosaurs disappear; mammals spread and diversify	Flowering plants (angiosperms) rise toward dominance

WORKED EXAMPLE — READING THE TIMELINE BACKWARDS

Question: a rock layer holds a jawless-fish fossil, and a layer above it holds a dinosaur bone. Which is older, and is the order consistent with the timeline?

- **Step 1:** deeper layer = older, so the jawless fish lived earlier.
- **Step 2:** check the dates — jawless fish appeared about **500 mya**; dinosaurs dominated from about **230 mya**. The order (fish below, dinosaur above) is consistent.
- **Step 3:** a trick variant — "could the fish be **younger** than the dinosaurs?" No: 500 mya comes **before** 230 mya in deep time. Bigger mya number = older.

Train the reflex: convert every date to "years before present", and the larger number is always the earlier event.

ANALOGY — EARTH'S HISTORY AS A CALENDAR YEAR

Compress 4.5 billion years into one year of 365 days, so each day carries about 12 million years. Earth forms on 1 January; the first cells appear in late February; jawless fish arrive in mid-November; dinosaurs rule from about mid-December; they vanish around 26 December; mammals have a quiet Boxing Day start; "Lucy" walks at about 5 pm on 31 December; Homo sapiens appears about 9 minutes before midnight; and agriculture begins in the final minute. The mapping is arithmetic, not decoration — divide each date by 12 million years per day and place it on the calendar — and it teaches the honest lesson: our kind is a newcomer.

Origin and Evolution of Man

Every family has an album, and ours spans millions of years. Two ideas must be nailed down before the names. First, humans and apes **share ancestors** — we did not descend from any ape alive today; both sides branched from common stock. Second, the story is **not a ladder of effort** with one species handing over to the next: several human-like species lived at the same time, so the family tree is a **bush of relatives**, some of whom were simply not our ancestors.

The ladder, rung by rung

Species	When and where	What marks each rung
Dryopithecus and Ramapithecus	About 15 mya; Africa, Asia, Europe	Ape-like, hairy, walked like gorillas and chimpanzees; Ramapithecus carries the first man-like hints
Australopithecus	3–4 mya; East African grasslands	"Lucy"; walked upright, about 2–4 feet tall; hunted with stone weapons but essentially ate fruit
Homo habilis	About 2 mya; Africa	Brain about 650–800 cc; the first toolmaker — "handy man"
Homo erectus	About 1.5 mya	Larger brain, about 900 cc; almost certainly ate meat
Neanderthal man	100,000–40,000 years ago; Near East, Central Asia, Europe	Brain about 1400 cc; used hides to protect the body; buried their dead
Homo sapiens	Arose in Africa about 75,000–100,000 years ago	Spread across the continents; cave art about 18,000 years ago; agriculture about 10,000 years ago; settlements and civilisations follow

- **Dryopithecus and Ramapithecus:** the oldest pair in the syllabus ladder — mostly ape-like, but Ramapithecus shows the first man-like features.
- **Australopithecus:** the famous "Lucy" skeleton; upright walking came before big brains — bipedalism was the first step of the human line.
- **Homo habilis:** the name means "handy man"; stone tools are its signature.
- **Homo erectus:** a bigger brain (about 900 cc) and a meat-eating diet.
- **Neanderthal man:** a large brain (about 1400 cc), hides for clothing, and the touching habit of **burying their dead**.
- **Homo sapiens:** arose in Africa, moved across continents during the ice age, painted caves about 18,000 years ago, began agriculture about 10,000 years ago, and built settlements and civilisations from there.

Reading the ladder correctly

- **Brain size grew in steps:** roughly 650–800 cc (habilis) → 900 cc (erectus) → 1400 cc (Neanderthal) → modern sapiens near that same range.
- **Traits did not arrive together:** upright walking, tool-making, big brains and farming each appeared at a different rung — no single jump made a human.
- **Not a ladder of effort — a bush of relatives:** several species coexisted, and in the NCERT ladder **Neanderthal man is a close relative branch, not a direct ancestor** of Homo sapiens.
- **Dates to carry:** 15 mya (ape-like pair) → 3–4 mya (Lucy) → 2 mya (tools) → 1.5 mya (erectus) → 100,000–40,000 years ago (Neanderthal) → 75,000–100,000 years ago (sapiens) → 10,000 years ago (agriculture).

ANALOGY — A FAMILY PHOTO WALL, NOT A RELAY RACE

Picture the evolution of man as a photo wall rather than a relay race. The relay picture (one runner passing a baton to the next, each better than the last) is what students imagine — and it is wrong. The photo wall shows a big extended family: cousins who lived at the same time, aunts and uncles whose lines ended, and one thread that leads to us. Neanderthal is the tall cousin in the corner — related, successful in his own time, not an ancestor standing "before" us. That is what "a bush of relatives" means.

Key Facts & Terms Cheatsheet

KEY FACTS & TERMS CHEATSHEET — EVOLUTION

Term / fact	What it means	Example / number to carry
Miller–Urey experiment (1953)	Early-air gases + electric spark built life's building blocks	$CH_4 + NH_3 + H_2$ + water vapour at about 800°C → amino acids
Oparin–Haldane idea	Chemical evolution — life assembled step by step from non-living molecules	Gases → organic molecules → first cells
RNA world	RNA was the first genetic material — it stores information and catalyses reactions	First cells about 3.8 bya; all early cells prokaryotes
What selection needs	Heritable variation + struggle for existence + survival and reproduction of the fittest + branching descent	Fittest = most surviving offspring, not strongest

Term / fact	What it means	Example / number to carry
Homologous organs	Same basic structure and origin, different functions → divergent evolution	Whale flipper, bat wing, cheetah leg, human arm; Bougainvillea thorn vs Cucurbita tendril
Analogous organs	Different structures and origins, same function → convergent evolution	Butterfly and bird wings; octopus and mammal eyes; penguin and dolphin flippers
Adaptive radiation	One ancestor radiates into many forms in new niches	Darwin's finches (seed → insect → woodpecker-like beaks); Australian marsupials vs placental look-alikes
Hardy–Weinberg equation	$p^2 + 2pq + q^2 = 1$ and $p + q = 1$; p^2 = AA, $2pq$ = Aa, q^2 = aa	$p = 0.7$, $q = 0.3$ → AA 49%, Aa 42%, aa 9%
Five disturbing factors	Gene migration, genetic drift, mutation, recombination, natural selection	Drift is strongest in small populations (founder effect, bottleneck)
Three selection modes	Stabilising keeps the average; directional shifts it; disruptive favours both extremes	Birth weight / drought beaks / seed-size beaks
Industrial melanism	Soot-darkened trunks flipped which moth form survived	Biston betularia: white common before industry, dark after
Timeline anchors	Earth 4.5 bya; first cells 3.8 bya; invertebrates 2000 mya; jawless fish 500 mya; ferns and amphibians 350 mya; dinosaurs 230 → gone 65 mya	Dinosaurs dominated for over 160 million years
Human ladder in order	Dryopithecus/Ramapithecus → Australopithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens	15 mya → 3–4 mya → 2 mya → 1.5 mya → 100,000–40,000 ya → 75,000–100,000 ya

Common Errors and Special Cases

COMMON ERRORS & SPECIAL CASES

The mistake students make	What is right	One-line why / the trap
Sorting organs by **job** ("both fly, so same")	Test the **structure and origin** first: same plan = homologous even when jobs differ	Bird wing and butterfly wing both fly — yet they are analogous, not homologous
Pairing "convergent = homologous"	Convergent evolution produces **analogous** organs; divergent evolution produces **homologous** organs	The pair flips under pressure — rehearse it until it is automatic
Treating genetic drift as a form of selection	Drift is a **random** change in allele frequency, strongest in small populations	No allele is "better" — chance alone can raise, fix or erase one
Applying Hardy–Weinberg to any population blindly	It holds only with a large population, random mating, and **no** selection, mutation, migration or drift	When those forces act, frequencies shift — and that shift **is** evolution
Calling $2pq$ the dominant phenotype	$2pq$ is only the heterozygote frequency; the dominant phenotype is $p^2 + 2pq$	In the $p = 0.7$ example: carriers 42%, but dominant look = 49% + 42% = 91%
Saying "humans descended from Neanderthals"	In the NCERT ladder Neanderthal is a close relative branch, not a direct ancestor	Several hominid species coexisted — the lineage is a bush, not a ladder
"Miller made life in a flask"	Miller made **amino acids** — the building blocks — not living cells	Cells also need membranes, metabolism and heredity, which no flask experiment produced

Three checking habits before you move on

- **The two-question habit:** for any organ pair, ask "same plan?" **before** "same job?" — the plan answer decides homology, the job answer only confirms the analogy case.
- **The assumption habit:** before any Hardy–Weinberg computation, list what must be absent — selection, mutation, migration, drift — and confirm mating is random; if a factor is acting, frequencies are shifting and that shift is the answer.

- **The order habit:** for timeline and human-ladder questions, write the rungs as a rough line first ("cells → fish → ferns → amphibians → dinosaurs → mammals"; "Lucy → tools → erectus → Neanderthal → sapiens"), then attach the dates — order questions break when dates are memorised without the line.

The Whole Chapter on One Page

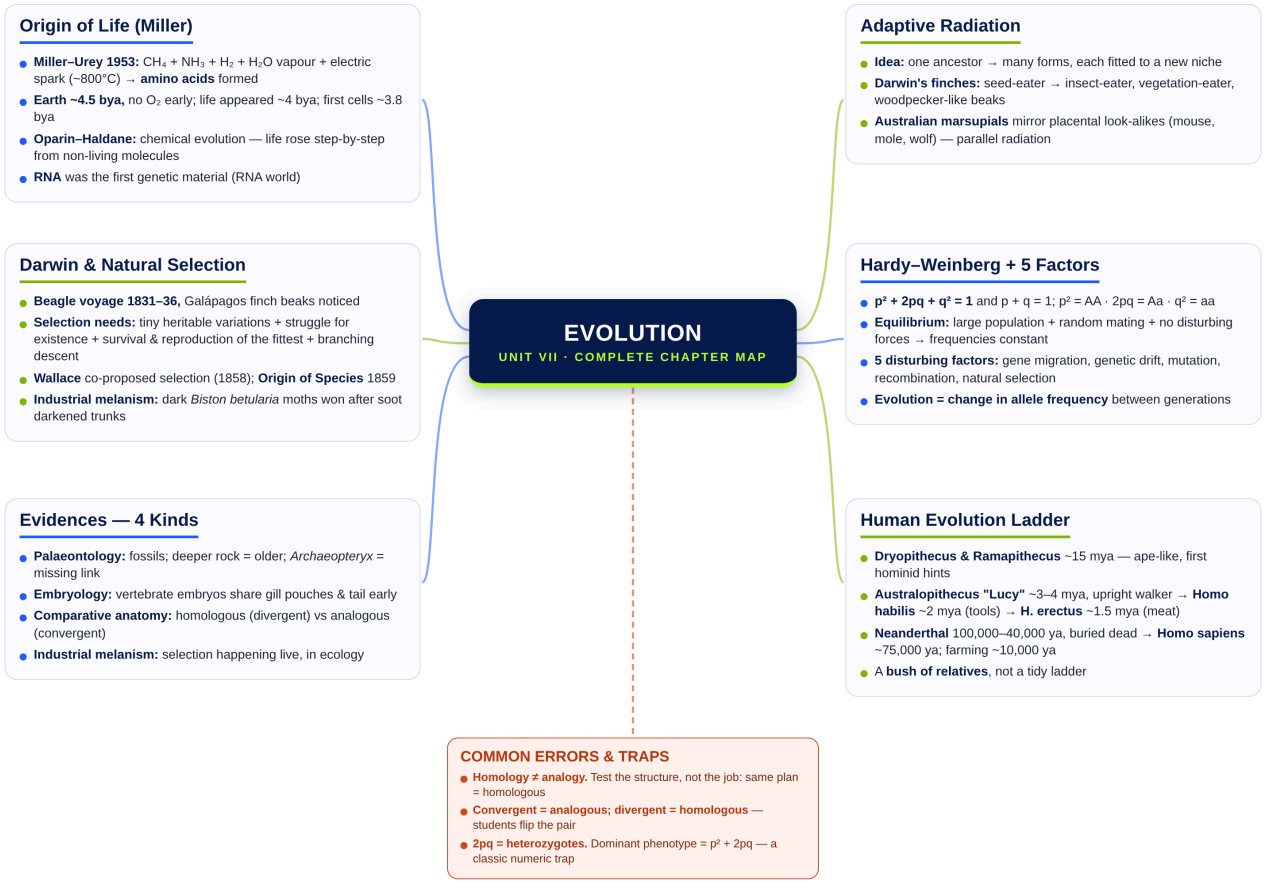


Figure 2 — Whole chapter on one page: origin of life, Darwin's mechanism, the four evidence kinds, adaptive radiation, Hardy–Weinberg with its five factors, and the human ladder, with the three traps that catch most students.

Practice Questions

Q1. In Miller's 1953 experiment, the electric spark was passed through a sealed flask containing:

(A) CO_2, O_2, N_2 and water vapour
(B) CH_4, NH_3, H_2 and water vapour
(C) CH_4, O_2, N_2 and water vapour
(D) CO_2, NH_3, O_2 and hydrogen gas

Q2. During industrial melanism, dark-winged peppered moths became the common form in soot-covered areas because they:

(A) ate more of the soot-covered lichen
(B) laid more eggs than the white moths
(C) were better camouflaged on dark trunks, so predators caught fewer
(D) changed their wing colour to dark during their own lifetime

Q3. Which of the following pairs shows **homologous** organs?

(A) Thorn of Bougainvillea and tendril of Cucurbita
(B) Wings of a butterfly and a bird
(C) Flippers of a penguin and a dolphin
(D) Eyes of an octopus and a mammal

Q4. Convergent evolution produces:

(A) analogous structures
(B) homologous structures
(C) vestigial organs
(D) missing links

Q5. In a population at Hardy–Weinberg equilibrium, $p = 0.8$ and $q = 0.2$. The frequency of heterozygous individuals is:

(A) 0.16
(B) 0.32
(C) 0.64
(D) 0.04

Q6. In a population at Hardy–Weinberg equilibrium, 16% of the individuals show a recessive trait. The frequency of heterozygous carriers is:

(A) 16%
(B) 24%
(C) 48%
(D) 96%

Q7. A few individuals colonise a remote island, and the new population's allele mix differs from the mainland's purely by chance. This is an example of:

(A) gene migration
(B) mutation pressure
(C) disruptive selection
(D) genetic drift (founder effect)

Q8. After a drought, only large hard seeds remain, and over many generations finches with larger beaks survive and reproduce more. This shift is:

(A) stabilising selection
(B) directional selection
(C) disruptive selection
(D) genetic drift

Q9. Which option places these in the correct order of appearance?

(A) Homo habilis → Australopithecus → Homo erectus → Homo sapiens
(B) Homo erectus → Homo habilis → Australopithecus → Homo sapiens
(C) Australopithecus → Homo habilis → Homo erectus → Homo sapiens
(D) Australopithecus → Homo erectus → Homo habilis → Homo sapiens

Q10. Dinosaurs dominated the Earth from about one date and disappeared by another. The correct pair of dates (mya) is:

(A) 500 and 200 **(B)** 350 and 150
(C) 150 and 65 **(D)** 230 and 65

Solutions

Sol. Q1. Miller rebuilt the early atmosphere: CH_4, NH_3, H_2 and water vapour — **no oxygen**, kept at about 800°C with an electric spark for lightning. The mix in option B is exactly his recipe; every other option slips in oxygen or drops a key gas.

Answer: (B)

Sol. Q2. The moths rest on trunks by day, where birds hunt by sight. Soot darkened the trunks, so dark wings were the better camouflage — predators caught fewer dark moths, which then left more offspring. Individual moths never change colour (option D is the classic trap).

Answer: (C)

Sol. Q3. Thorn and tendril are both **modified branches** — same plan and origin, different jobs — so they are homologous. Butterfly and bird wings, penguin and dolphin flippers, and octopus and mammal eyes all do the same job from unrelated starting parts: analogous.

Answer: (A)

Sol. Q4. Convergent evolution presses **different** structures toward the **same** function — which is the definition of analogous structures. Homologous structures belong to divergent evolution, where the same plan takes different jobs.

Answer: (A)

Sol. Q5. Heterozygotes are $2pq$: $2 \times 0.8 \times 0.2 = \mathbf{0.32}$, that is 32%. Option A (0.16) is p^2 or q^2; option C (0.64) is p^2 alone — the trap is grabbing a single squared term instead of the cross term.

Answer: (B)

Sol. Q6. Recessive individuals are q^2: $q^2 = 0.16$, so $q = \sqrt{0.16} = 0.4$. Then $p = 1 - 0.4 = 0.6$. Heterozygotes: $2pq = 2 \times 0.6 \times 0.4 = 0.48$ — **0.48**, that is 48% of individuals, carry the allele without showing the trait.

Answer: (C)

Sol. Q7. The colonists' alleles are a **random sample** of the mainland pool — chance, not fitness, set the new frequencies. That is the **founder effect**, a form of genetic drift. Gene migration means alleles moving between existing populations; mutation adds new alleles.

Answer: (D)

Sol. Q8. Selection now favours **one extreme** (larger beaks), so the population average moves in that direction — directional selection. Stabilising would favour the average beak; disruptive would favour both extremes at once.

Answer: (B)

Sol. Q9. The ladder runs Australopithecus (3–4 mya) → Homo habilis (2 mya) → Homo erectus (1.5 mya) → Homo sapiens. Options A, B and D all swap habilis and erectus or start with the wrong rung — the tool-maker came before erectus.

Answer: (C)

Sol. Q10. Dinosaurs rose about **230 mya** and disappeared about **65 mya** — a run of over 160 million years, ended suddenly. The other pairs mix up jawless fish (500 mya), ferns (350 mya) and mammal spread (after 65 mya).

Answer: (D)

60-Second Recall Drill

Cover the right column and say each answer aloud before checking it.

Cue	Say this
Miller's four gases	CH_4, NH_3, $H_2 + H_2O$ — spark, 1953, amino acids, no oxygen
Homologous →	Same plan, different jobs (forelimbs of whale, bat, cheetah, human) — divergent
Analogous →	Different plans, same job (butterfly and bird wings) — convergent
Adaptive radiation	One ancestor, many forms — Darwin's finches; Australian marsupials
Hardy–Weinberg	$p^2 + 2pq + q^2 = 1$; p^2 = AA, $2pq$ = Aa, q^2 = aa; $p + q = 1$
Five factors	Gene migration, genetic drift, mutation, recombination, natural selection
Genetic drift	Chance change in allele frequency; strongest in small populations
Three selection modes	Stabilising = average; directional = one extreme; disruptive = both
Human ladder	Australopithecus → Homo habilis → Homo erectus → Neanderthal → sapiens

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